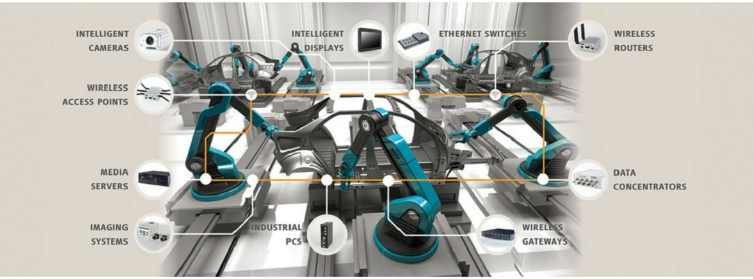
The Evolution Of Industrial Revolution—IIoT



Industry 4.0 (Image courtesy: www.eltec.de)

Vertica Asthana is a technical journalist at EFY

There was a catastrophic incidence of oil spillage from the oil rigs in Gulf of Mexico in April 2010. The drilling rig 'Deepwater Horizon' had spilled about 185 million gallons of oil into the sea, endangering breeds of marine creatures. Later, forensic reports proved that a puncture in the pipe due to oil and gas pressure had created the havoc. Had there been a smart automation system in place, the fault would have been detected at an early stage, hence avoiding the mishap.

This incident highlights the necessity of smart automation in industries. And as it is said "necessity is the mother of disruption," the dire need of a safe, secure and efficient industrial system gave birth to the Industrial Internet of Things or IIoT.

The IIoT roadmap

There was a time in the 18th century when industrial machines were powered by steam. Industry 2.0 was powered by electric energy and industry 3.0 went autonomous. Now we have industry 4.0, which runs on cyber-physical systems.

This paradigm shift of the industry norms from manual to smart automation is helping to remotely monitor and control every part of the industrial facility.

"Later, five considerations were taken for enabling IIoT—distributed intelligence, rapid connectivity, establishing open standards and systems, real-time context integration and autonomous production lines," says Narendra Sivalenka, senior manager-IoT team, Cyient.

Extracting data from all systems or extricating data from the main system and placing it into different machines in perfection is the distributed intelligence in the IIoT. This data is then connected with each other in a network after integration. Later, as per customer needs, this data is extracted in a smart way to avoid monotony of work, maximise throughput and minimise labour cost.

Technology behind the IIoT

Hardware components of the IIoT include sensors, RFID, condition monitoring,